

Poetics of Future Work: Blending Speculative Design with Artistic Methodology

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ABSTRACT

This case study presents how the mixing of speculative design with artistic methodology can contribute to the inquiry of technological potentialities in the future of work. The goal and belief are that technologies such as artificial intelligence can augment employee creativity and support their well-being at work. The co-design process followed an artistic approach and consisted of three cycles of labs, workshops and events during the span of one year to support professionals with non-technical background in the ideation and conceptualization of possible futures. The artistic approach consisted of different exploration perspectives of technology through the use of embodiment, artifacts and creation of speculative fictions. The research team that facilitated the labs was interdisciplinary and the participants were assembled from different partner organizations from industry and public sector. We share the learnings from this study attending to three different perspectives: our learnings from the facilitation of the artistic approach, our learnings from the future of work ideas and concepts developed by participants, and discussion of what these learnings can mean to design practitioners and the research community. Results indicate that embodiment and speculative fiction can create engagement among professionals that lack technical expertise and support them in collaborative exploration of alternative futures of work with novel and abstract technologies such as AI.

CCS CONCEPTS

• Human-centered computing; • Interaction design; • Interaction design process and methods; • Participatory design;

KEYWORDS

Artificial intelligence, speculative fiction, artistic methods, future of work

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1 INTRODUCTION

There is a growing consensus around the transformative and innovative power of new technologies such as Artificial Intelligence (AI) and Internet of Things (IoT), and the effect that these technologies will have on the future of work. A growing fear and uncertainty around the future of work is felt as AI is increasingly capable of performing more complex cognitive tasks. Dystopian projections of the future have been strengthened by the mainstream media debates, and increasingly popular sci-fi series and films. The fears of machines replacing humans are partially supported by the research [10, 15, 24], but also often biased and mixed with limited knowledge and understanding of AI and unrealistic expectations of its potential, assigning it almost magical powers. As the expert report by the Finnish Ministry of Economic Affairs and Employment [15] proposes, most jobs, at least in developed countries (like Northern Europe), will have to be partially re-defined in the future and people will need to re-skill to adjust to the technological development. However, it is not likely that the majority of jobs will be simply replaced by AI. New jobs will be also created and will demand a new set of skills, some of them not yet identified [15].

A more alarming challenge today is that technology has advanced more than people's readiness to absorb it. As a consequence, the main work to be done in the coming period is not in tech development itself but on the side of changing people's behaviors and facilitating a massive process of education, organizational transformation and re-skilling of workforce [15]. This transformation will put an increased pressure on organizations to find ways to proactively engage employees in discussing their future work. Both to raise their awareness and knowledge about AI and other digital technologies, but also transforming employees' fears of being replaced into exploration of new possibilities created by technology, with the purpose of augmenting their productivity, innovation and well-being.

As a way to address these challenges, one possibility is to engage people in collaborative design activities [21], where there is a tradition to design generative toolkits to support non-designers in creative processes, having as inspiration tools that introduce technology (e.g., IoT) to nonprogrammers [17, 19]. In this case-study, we introduce the research project named "DIGI Futures", which took place in 2019 and had two main goals. On one hand, it tried to shift the focus from technical aspects of digitalization and the application of new technologies to increase productivity, towards more human aspects of digitalizing work, such as increased balance, well-being and creativity at work. On the other hand, the focus was on developing an innovative methodology that would empower employees with non-technical backgrounds to be part of creating visions and exploring possible futures, reflecting on both potential challenges and opportunities of digitalizing work.

To fulfill the project goals, the project coordinator RISE formed an interdisciplinary research team that included experts from user experience, various technical domains (AI, IoT, virtual reality), innovation management, organizational studies, arts and philosophy. External partners participating in the project also represented a wide spectrum of work contexts, namely: industry (ABB, Electrolux, GodEl), public sector (Karolinska hospital and Nacka municipality), and civil society (creative hub and interdisciplinary community NAV Sweden). Participants from these organizations work in various organizational functions (innovation, IT, human resource management, sustainability, marketing, supply chain, etc.) and levels in organizational hierarchy, which enabled the research team to explore how people from various backgrounds can relate to the field of the future of work and enrich it with different perspectives. In terms of developing innovative methodology for empowering employees in the process, we took inspiration from the field of speculative design [8] but blended knowledge from this field with artistic methods and practices that we believed could increase engagement and ownership by project participants from industry and public sector.

2 METHODOLOGY

Based on the project goals and previous experiences of leading interdisciplinary projects around complex societal issues it was decided to use methods from arts and speculative design to: 1) increase project's creativity and innovation potential; 2) bridge differences between participants coming from a wide variety of backgrounds; 3) find fun and creative ways for people with non-technical backgrounds to get interested and engaged with abstract topics such as AI; 4) increase active participation and co-creation among project participants.

The project lasted one-year and was divided into three different cycles. Each cycle had a topic in focus that related future of work with a specific technology: Cycle 1, "Unboxing AI for employees"; Cycle 2, "IoT - empowering people, not just things"; and Cycle 3: "Virtual reality - the endless training center of tomorrow?" (see Figure 1). As the project evolved, it became clear that more than focusing on specific technology, participants from the different partner organizations wanted to focus on shared challenges related to the future of work. This is why participants decided to refocus Cycle 3 towards exploring future jobs and the needed re-skilling due to increased digitalization and automation of work.

Before the cycles started there was a kick-off workshop where project partners and the research team got to know each other, agreed about how they wanted to work together, and co-created a project plan. Every cycle was initiated with a 2-day interdisciplinary lab where the research team and 1-2 representatives from each partner organization would explore different possible futures related to the topic in focus. The design process of interdisciplinary labs was emergent and changed in time, adjusting to the needs and learnings in the project. Labs were designed by a core team of facilitators, consisting of the project leader who is a senior researcher in innovation and design and two artists: one who is a musician and visual artist and the other who is a choreographer and performing artist. Facilitators collaborated with a tech expert on the topic of each lab during the lab preparation. In general, day one of every lab

was devoted to opening up new perspectives on the topic in focus and on mapping trends, questions and concepts relevant for that topic (see Figure 1). In line with the idea of speculative design, the focus was on asking questions that would encourage enquiry into possible issues of digitalization and stimulate discussion around them [1]. Day two was the day of developing different possible futures centered around questions prioritized on day one. Participants developed different speculative fictions and artistic methods were used in the process to encourage play, exploration, and a collective re-imagining of a future world in line with the idea of seriously silly design fiction by Blythe et al. [3]. After each lab, there were workshops that aimed at transferring and contextualizing knowledge from the lab into the partner organizations in a more compact form (2-4 hours).

In the end of each cycle there was a public event where knowledge created in the cycle was shared with a larger audience. Apart from round table discussions, new formats of engaging audience were tested in these events. For example, through a live concert at the Electrolux headquarters where original music, inspired by the speculative fictions from the project, was written and performed (see Figure 2). Another example of an innovative format used in events were interactive art exhibitions. The audience (which included but was not limited to project participants) was usually invited to co-create artifacts from the future which were exhibited in the end of the event. For the last event, visual artworks representing characters from selected speculative fictions produced in the labs were exhibited and accompanied by audio stories and interactive animations. For each event, a special scenography was built to support the topic of the event and create a more intimate and immersive environment. For example, for the event of IoT and the future of work, different objects were floating from the ceiling among soft fabric waves and participants were invited to choose two objects and marry them into a smart object of the future. After the three cycles, an evaluation workshop took place to reflect on the main learnings and outcomes and agree about next steps in the research process.

2.1 Examples of Methods Used in the Research Process

2.1.1 Embodiment Methods. Different types of embodiment methods inspired by contemporary performing arts practices were used throughout the project. The project leader specifically built on her experiences of working with contemporary dance choreographers during the last ten years, applying methods, practices and theory from contemporary dance in organizational context [4, 7, 12–14, 16, 25, 26].

Every lab started with a warm-up session where movement exercises were applied to enable participants to arrive into the physical space and into the group, to connect with the topic of exploration and to open up their imagination. For example, in one exercise, participants were to embody an AI with a superpower they would like to augment themselves with in their future work. Suddenly, the room was filled with AIs, embodied in a variety of ways, meeting and interacting with each other (see Figure 3). This type of exercises engaged participants in more kinesthetic ways and increased playfulness and creativity in the group.

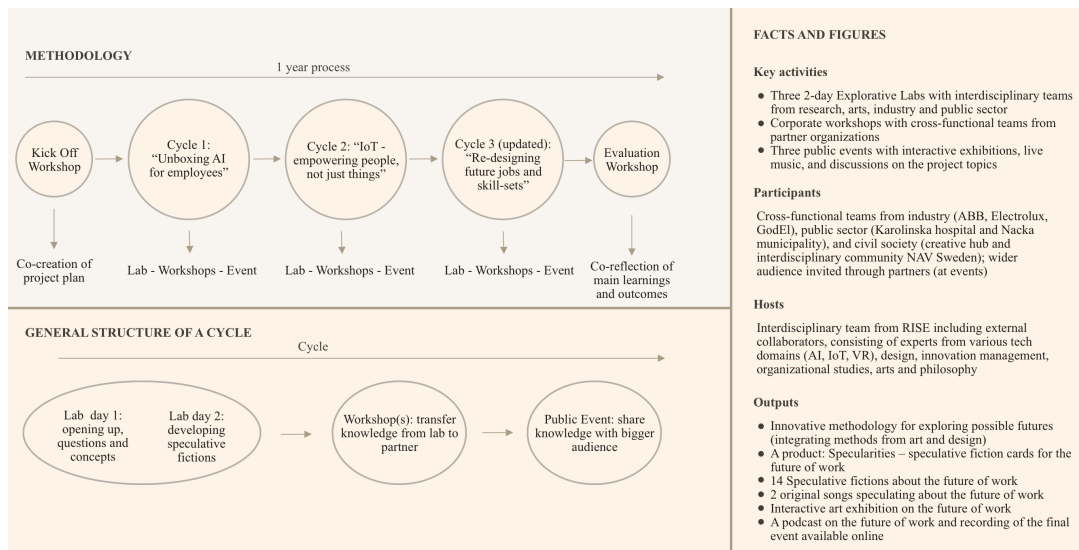


Figure 1: On the left, structure of the project divided into three cycles. Each cycle included a lab, workshop(s) and public event. On the right, facts and figures of the research project.



Figure 2: Pictures taken during different labs, workshops and events. From left to right, in the first row: Lab 2, group discussion on day 1; Lab 2, participants working on speculative fictions on day 2. In the second row: Workshop 1 at ABB, participants work on speculative fictions; Workshop 3 with external guests, participants playing with speculative fiction cards. Picture to the right: Event at the Electrolux headquarters in Stockholm, retro futurist song performed by Dragana Zarevska and Andi Kravljaca.

Other types of movement exercises would be used to create embodied group sculptures, representing both the current ways of relating to technology at work and possible future visions. In between the two, participants would try out different possible scenarios of trajectories between the present and future visions through movement. In this way, they experienced how relations between technology and humans could evolve not only on a cognitive level but also feeling them through body and emotion, inscribing memories of possible futures in more visceral ways.

The embodiment methods used in this project emphasized the live interaction between participants that was improvised in the moment and focused on participants' experience. The enactments were relating to participants' real life experiences and daily situations at work, though they projected them into possible futures. In that sense they shared some elements with Speculative Enactments [9], but at the same time they were also very different. In Speculative Enactments, there is a lot of design work involved in the staging of the enactment, while in our case participants themselves enacted



Figure 3: On the left, warm-up session of Lab 1, participants meeting different AIs embodied by humans. On the right, participants embodying possible visions of the future of work with technology in Lab 3.



Figure 4: From left to right: Lab 1, Artifacts representing concepts and questions participants brought with them to the lab; Lab 2, “System of things”, a collection of artifacts representing smart office objects of the future; Lab 2, Interactive exercise where participants moved smart office objects of the future, exploring possible relations between them; Event 3, Artifacts representing Poetics of Future Work made by event participants.

their ideas about possible futures in the moment, as they were arising, and without any help of pre-made design props that would stage the whole experience. In comparison to User Enactments [20], the enactments were not scripted experiences with intentionally designed dialogues and encounters between participants. Participants would themselves create ideas and visions about possible futures and explore them on the spot through improvisation and movement with the help of facilitators.

2.1.2 Exploring through artifacts. Artifacts were also used in different ways during the research process. Participants were often invited to bring objects to the interdisciplinary labs that they personally related to the topics of exploration. For example, as a preparation for Lab 1 on AI and the future of work, participants were asked to think of three questions they personally found relevant to investigate in the lab. They were also invited to bring three objects connected to the questions (see Figure 4). During the lab, these artifacts were used in various playful ways to support the process of exploration, to represent questions explored in a physical way, to support storytelling and communication of developed ideas in the lab, and as an interaction tool between participants. In the second lab on IoT and the future of work, participants were invited to bring with them an object that for them symbolically represents a smart office object of the future. “A system of things” with various smart office objects of the future was then built collaboratively by participants. Relations between these future artifacts and people were

explored through interactive embodiment exercises. In all three public events, participants were invited to also make the future work artifacts, creating stories and characters around them and then co-creating a shared exhibition.

Compared to more established approaches from speculative design [8], these future artifacts were not carefully designed by designers and exhibited in museum/gallery environments, but rather improvised by project participants through fast prototyping. This fast prototyping process was used as a tool to materialize speculative ideas and stories about possible futures in a physical way, to help participants build the world of the future with their own hands. During one of the labs, participants visited a speculative design exhibition at the Swedish Centre for Architecture and Design in Stockholm, titled “The Future Starts Now”, using this experience an inspiration for developing their own speculative fictions.

2.1.3 Specularities. One of the main methods that we used throughout the process was inviting participants to co-create speculative fiction stories about possible futures of working with new technologies, such as AI, IoT, and VR. These stories were built around characters who represented various types of workers of the future and investigated both challenges and opportunities of integrating new technologies in the future of work. The idea, in line with speculative design, was not to focus on short-term solutions but rather provide a longer-term and more critical approach to technological development [1].

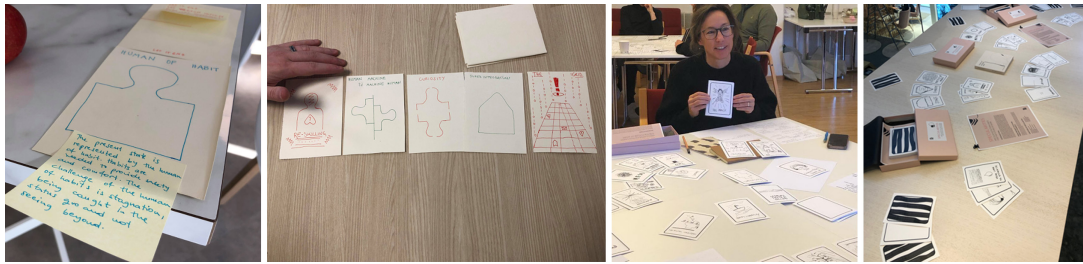


Figure 5: From left to right: Lab 3, example of a Specularity card made by a participant; Lab 3, a set of Specularities chosen to represent a speculative fiction; Workshop 3, participant playing with existing set of Specularities; Event 3, boxes with Specularities and instructions exhibited for the audience.

In the last lab, participants decided they wanted to explore how their own jobs might change in the future and what kind of transformation journeys, including re-skilling, they might go through to arrive to those futures. The team of facilitators decided to take tarot cards and Italo Calvino's novel *The Castle of Crossed Destinies* as an inspiration. Calvino's novel plays with the idea of storytelling without words, using images on tarot as a machine for telling an infinite number of stories [22]. Different writers have used tarot cards as an inspiration source for storytelling since the Italian and European Renaissance, where tarot has appeared in various speculative fiction works [15, 23]. Tarot has inspired also other card decks, such as *Intùiti Creative Cards* and *Interaction Tarot*, which is used as an interactive tool to support the early stages of design [6]. In this line of thought, we invited participants to create their own deck of cards, which would through visual symbols, their titles, and short descriptions inspire players to make their own associations and interpretations, supporting them to develop speculative stories about the future of work. We call these cards *Specularities*.

In order to embark on a journey of re-designing their future jobs, participants were first invited on an explorative visioning journey. They were asked to reflect on questions such as: How will your future work environment look like? What could be your job (title) in the future? What would you like to see yourself doing in the future? How could technology support and augment you? What might be your challenges in this future work landscape? How will your relationship and interaction with technology change? Based on these personal reflections and visions, each participant was then asked to create three cards. One card would symbolize their future of work vision, one would represent their current work situation, and one would symbolize the journey they would need to go through to move from the present into the future vision. Each card had a visual symbol, a title, and a short description of what it represents (see Figure 5).

Participants in the lab were then invited to play with the cards and first use them to improvise various speculative fictions around different invented characters from imaginary landscapes of future of work. The stories always followed a classical story-telling structure and introduced the character of the story with the first card, the context with the second, the conflict with the third, the resolution with the fourth and the message/moral of the story with the last-fifth card.

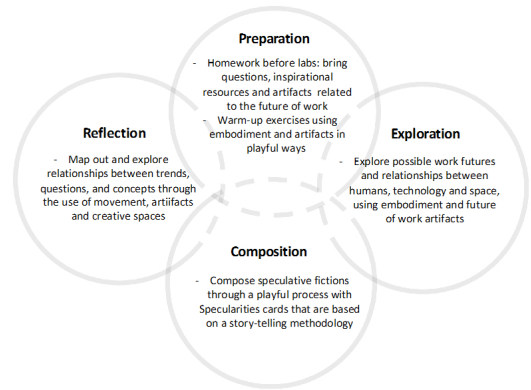


Figure 6: Iterative cycles between preparation/ tuning-in; exploration; reflection; and composition inspired by the model of innovative practice [4]

In the end of the process, each group developed a speculative fiction in which they chose one person from the group and created a story about them as a future worker, rethinking their job, their skillset, and their journey from the present into the future.

After Lab 3, Specularities created by participants in the lab were “productified”. They were printed in a black & white version together with empty Specularities that could be created by future players and packaged in boxes together with instructions on how to use them. The “productified” version was then tested in a shorter format in Workshop 3 with the idea that other companies could use a similar workshop structure in the future. Figure 5 depicts how cards were used in a workshop set-up and how the packaging with instructions looked like.

Figure 6 showcases key aspects of the creative process that were used when designing labs and workshops in the research process, with examples of methods used in each part of the process. This design of the process is based on research about innovative practices and processes inspired by artistic methods [4].

2.2 Content-Related Findings and Results

The researchers selected the Affinity Diagram process [11] to analyze the fourteen different speculative fictions from Cycles 1, 2 and 3 written by participants. The affinity diagram was adapted

to analyze the stories and followed three stages, from highlighting different topics appearing in the stories, to discussing the relevance of identified topics and to the clustering of the topics into higher level themes. The result of the Affinity Diagram Process was a selection of themes that reflected different dimensions and qualities considered by participants in the applicability of AI, IoT and VR in their future of work. As a general reflection from the analysis, AI took a more relevant role than IoT and VR in the technological aspects of the fictions. This is reflected in the lessons learned, which are summarized as follows:

AI Work Augmenters. In the fictions, AI is often depicted as a work assistant in different forms, namely robots, and smart objects that can augment human workers in various ways. For example, by summarizing messages, representing people in meetings, giving them feedback when practicing a presentation, scheduling time for specific activities and personalizing physical work environments. In several cases AI is associated with human qualities, assigning it the capacity to feel and be empathic, and even referring to it as an independent employee or CEO.

Often, fictions depicted uses-cases where AI could eliminate or reduce routine aspects of work and help humans redistribute time in a more meaningful way that would increase their creativity and well-being. For example, by allowing them to focus more on the work they want to do, spending more time on creative and strategic tasks. Or by helping them enhance human needs and qualities, supporting them in scheduling work with the purpose of finding a better tempo and avoid stress and overworking. Social aspects of work were also explored in different ways, such as AI supporting humans to achieve a healthier balance between social life and work, so humans can spend more time socializing with colleagues and their family.

AI assistants in the speculative fictions were also acting as coaches, helping humans to be more reflective and to live more according to their values. By using AI to process biological data and other sources of information that people might not consciously be aware of, the AI assistants could augment people in ways that was previously not possible.

Imbalance of power. One of the main components in speculative fictions is that the characters need to face also potential negative effects of incorporating AI technologies into work. Introduction of AI could create an imbalance of power at work, for example AI misinterpreting requests of people and going too far in either taking on too much responsibility or manipulating people. Another challenge could be to treat them like machines, over scheduling their work, de-humanizing the person and making them feel like a robot. An example is when the AI assistant “has taken her instructions a bit too literally, cancelling many meetings to create more time for her [...]” and that leads to the protagonist feeling “powerless an even more lost than before, like she’s acting on some autopilot [...]”.

Data Privacy. Participants shared various concerns about data privacy and ownership of data in the context of future work. An example is when an AI decides to record an interview without the consent of people because the AI has predicted that it will be consented: the AI “calculated that we will agree to record the interview”. These considerations raise the issues of mistrust in AI

and potential challenges with transparency of what technology is doing and for whose benefit.

AI inventions, services and companies. Participants imagined different AI-inventions in the form of products, services and companies in the speculative fictions. Examples of some of these inventions are: HUMA, a “(human-machine) therapy where the therapist helps mediate conversations between humans and smart machines in order to help them find the right synergies in their shared lives”; Mamba, a “figure that recorded meetings, collected data about different people in the office, and could suggest customized games and creative exercises to bring more playfulness and creativity to meetings”; the service company GoodR, which “provides personalized continuity through smart offices and spaces”, so no matter where people go, they can access the same type of office environment; and the Relax Womb BIBI, which is a room that “measures user’s pulse, pressure, level of stiffness and hydration, helping the user relax by offering personalized massages, guided mindfulness exercises, soft music, nice smells and enclosed cozy environment to lay in”.

2.3 Method-Related Findings and Results

The blending of artistic methods with ideas from speculative design, and the combination of embodiment, future artifacts and speculative fiction enabled participants to open up their imagination, immerse fully in the process of exploring possible futures, and release themselves of the limitations of their existing work environment. It activated their creative abilities in new ways and made them actively engaged, owning the outcomes of the process. It also supported non-tech people to explore their possible futures with technology, making something as abstract as AI more accessible, relatable and fun to engage with due to the playfulness applied in the method of exploration.

The case-study presented here shows that methods from speculative design and arts, which might be otherwise predominantly applied in the university set-up with students [18], can be also adopted by industry and public sector if adjusted to their needs. It is helpful to identify change agents who are interested to drive such explorative future work in organizations. They will be most probably the ones who will be prepared to participate in more time-consuming activities (in our case two-day labs), who will engage other colleagues, and who will also contextualize and further disseminate knowledge in their organizations through shorter formats (in our case 1-4h workshops).

The importance of having several learning cycles during a longer time period was also identified. On one hand, this gave participants enough time to learn about the methodology developed in the project and adapt it to the needs of their own organizations. And secondly, it allowed process facilitators to iterate and improve the methodology for exploring possible futures and developing speculative fictions during the project. As a consequence, one of the outcomes was a product: speculative fiction cards for the future of work called “Specularities”.

An appreciated feature of Specularities was that even once the cards were printed and packaged with instructions, the original drawings by participants from industry and public sector were

intentionally kept and not replaced by professional drawings. Participants appreciated this level of fidelity because the “imperfection” made cards more relatable and encouraged new players to feel the confidence to create their own Specularities. Some of the people who played with Specularities in the DIGI Futures project decided to continue using Specularities in different ways in workshops they facilitated after the project.

One challenge with methods tested in this case-study is that the output is largely dependent on participants’ engagement and input. When participants are lacking expertise (either in tech or in creative skills, like story-telling), the quality of output, for example in this case speculative fictions, can be impacted. Another challenge is that artistic methods can increase creativity in the process, but for some people from business they might be experienced as out of their comfort zone. This is why it is important that facilitators can adapt methodology to specific needs of each group. In the project described here, it was experienced that the core group of participants from industry and public sector who were part of the project throughout the process and were often change agents in their organizations were more open to experimental methods than the average participants in the workshops that more widely represented employees from participating organizations. A third potential hinder to wider adoption of the methods tested in this project across an organization is that most companies have quite a short-term performance focus. Exploring possible futures is therefore outside of the scope of most employees’ immediate work reality so unless they can connect this type of work to concrete challenges they have at work and see that speculative fictions might be able to provide innovative ideas or solutions to them, their interest and engagement might be limited.

2.4 Research Findings and Lessons Learned

One observation was that participants’ attitudes towards digital technologies were in general ambivalent, merging enthusiasm with anxiety and fear. Over time, the feeling was that the more and longer participants were engaged in the process and the more knowledge they gained, the more they started to see positive opportunities of using digital technologies to augment themselves and improve their well-being, creativity and even happiness. Giving participants the power to create their own visions of the future of work gave them the feeling that they can be active co-designers of the future technology, impacting their life and work, instead of being just passive consumers manipulated and negatively affected by technology.

On the other hand, the limitation of using methods speculating about the future was that it sometimes felt too detached from the current reality of participants. Although participants found the process inspiring, engaging, and educational, there was a question of how to use the knowledge created in the project to change and impact their current work. This is why in the future, explorative methods applied in the project could be combined with development and testing of smaller-scale technical prototypes that could be a first step towards realizing some of the ideas from speculative fictions in the current work reality of participants.

3 DISCUSSION

Compared to speculative design, design fiction and speculative enactments [2, 8, 9], the main difference in approach applied in this project was that it was not designers or artists who developed speculative artifacts, fictions, and enactments, but participants from industry and public sector together with researchers, who acted not as experts but rather as equal participants in the exploration process. If in design fiction, the world-building refers to the design of a context in which a ‘diegetic prototype’ (the speculative artifact) is designed by designers [2], and in speculative enactments much of the design work involves the staging of the enactment by designers [9], in our case participants with non-technical backgrounds were given a transformative power to feel they are not invited into a future speculation but are rather creating it themselves. Through this live co-creation and direct experience of possible futures through enactment, memories from the future were inscribed in participants’ bodies, resonating much closer than if they were passive visitors in a museum. As Elsdon et al. [9] propose “the essence of our critique is that we should be more ambitious in how participants can become meaningfully involved with speculation, and in Candy’s [5] terms, utilize the ‘continuum of human experience’”.

4 CONCLUSIONS AND FUTURE WORK

We learned that using embodiment, artifacts and speculative fiction can support professionals who lack computational expertise in the process of collaborative exploration of alternative futures with novel and abstract technologies such as AI. Participants were able to imagine and embody AI, conceptualize AI products and services, and officiate new marriage relationships between objects to create new smart products of the future. In this process, participants wrote speculative fictions that reflected on how these technologies could be integrated into the work settings, pondering on the positive and challenging consequences of these novel integrations. These reflections are depicted in the speculative fictions, in which the main characters struggled to find a balance when new technologies were introduced, due to redistributions of time and value for the companies, and misinterpretations of AI assistants and coaches, which could overwhelm the characters.

The main contributions of this project are on one hand showing how methods from speculative design and art can be tailored to facilitate future-oriented processes for professionals, and not only students. The case study also highlights the potential of using speculative design and artistic methods to help non-programmers ideating futures with complex technologies such as AI, IoT and VR. Experiences from the project emphasize the importance of live experience and active engagement of participants in the process, who do not only react to staged speculative prototypes or experiences but rather explore possible futures through improvisation and co-creation in the moment with support of researchers and experienced facilitators.

The next step in the research is to explore how to start a transformation process in participating organizations, using ideas from speculative fictions to prototype real-life (tech) solutions that can help organizations move from their current position towards their desirable vision of their future of work.

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